



**NOAA  
FISHERIES**

# TOR 2 - Catch

## Atlantic Sea Scallop Research Track Peer Review

April 21-24, 2025

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## Term of Reference 2

*Estimate catch from all sources including landings and discards. Describe the spatial and temporal distribution of landings, discards, and fishing effort. Characterize the uncertainty in these sources of data.*



Credit: Dvora Hart

Effort reduction  
over time

- 1982: Management first implemented – minimum mean meat counts
- 1984: Hague line dividing Georges Bank into US and Canadian EEZs
- 1994: Meat count regs eliminated. Limited access with no new permits, effort (days at sea and crew size) restrictions, 3” (76.2 mm) rings, Georges Bank closures in December
- 1995: 3.25” (82.55 mm) rings
- 1996: 3.5” (88.9 mm) rings
- 1997: Fishery declared overfished
- 1998: First Mid-Atlantic closures
- 1999-2000: First Georges Bank reopenings
- 2001: First Mid-Atlantic reopenings, fishery declared rebuilt
- 2004: Area management system established, 4” (101.6 mm) rings
- 2007-2008: ITQ General Category fishery (5.5% of total landings)
- 2011: Annual Catch Limit (ACL) management

| Year    | DAS   | GB Acc | MA Acc |
|---------|-------|--------|--------|
| 1982-93 | NA    | NA     | NA     |
| 1994    | 204   | 0      | 0      |
| 1995    | 182   | 0      | 0      |
| 1996    | 182   | 0      | 0      |
| 1997    | 164   | 0      | 0      |
| 1998    | 142   | 0      | 0      |
| 1999    | 120   | 13,608 | 0      |
| 2000    | 120   | 27,216 | 0      |
| 2001    | 120   | 0      | 23,133 |
| 2002    | 120   | 0      | 24,494 |
| 2003    | 120   | 0      | 28,577 |
| 2004    | 42    | 8,165  | 32,659 |
| 2005    | 40    | 16,329 | 24,494 |
| 2006    | 52    | 40,824 | 0      |
| 2007    | 51    | 16,329 | 24,494 |
| 2008    | 35    | 8,165  | 32,659 |
| 2009    | 42    | 8,165  | 32,659 |
| 2010    | 38    | 8,165  | 24,494 |
| 2011    | 32    | 16,329 | 16,329 |
| 2012    | 34    | 20,412 | 12,247 |
| 2013    | 33    | 7,502  | 3,787  |
| 2014    | 31    | 5,210  | 5,210  |
| 2015    | 30.86 | 0      | 23,133 |
| 2016    | 34.55 | 0      | 23,133 |
| 2017    | 30.41 | 16,329 | 16,329 |
| 2018    | 24    | 32,659 | 16,329 |
| 2019    | 24    | 32,659 | 24,494 |
| 2020    | 24    | 24,494 | 16,329 |
| 2021    | 24    | 24,494 | 8,165  |
| 2022    | 24    | 20,412 | 0      |
| 2023    | 20    | 10,886 | 5,443  |

# Overview

- Typical limited access vessel is 25-35 m in length
- Most tow two toothless New Bedford-style dredges, typically 15' (4.57 m) wide with 4" rings, but there are "small dredge" permits which can tow only one 10.5' (3.2 m) dredge. A few vessels use trawls in the Mid-Atlantic.
- Primary port is New Bedford, MA (75% of landings in 2023)

## Limited access vessels

- Trips typically 6-14 days in length
- Allocated area-specific trips with a trip limit (essentially an IFQ) into recently reopened "access areas"
- Also receive a number of days-at-sea (DAS) to be used in "open" areas outside of closed or access areas



## General category vessels

- 'Day boat' vessels, restricted to 600-800 lbs/day plus ITQs (5.5% of fishery)

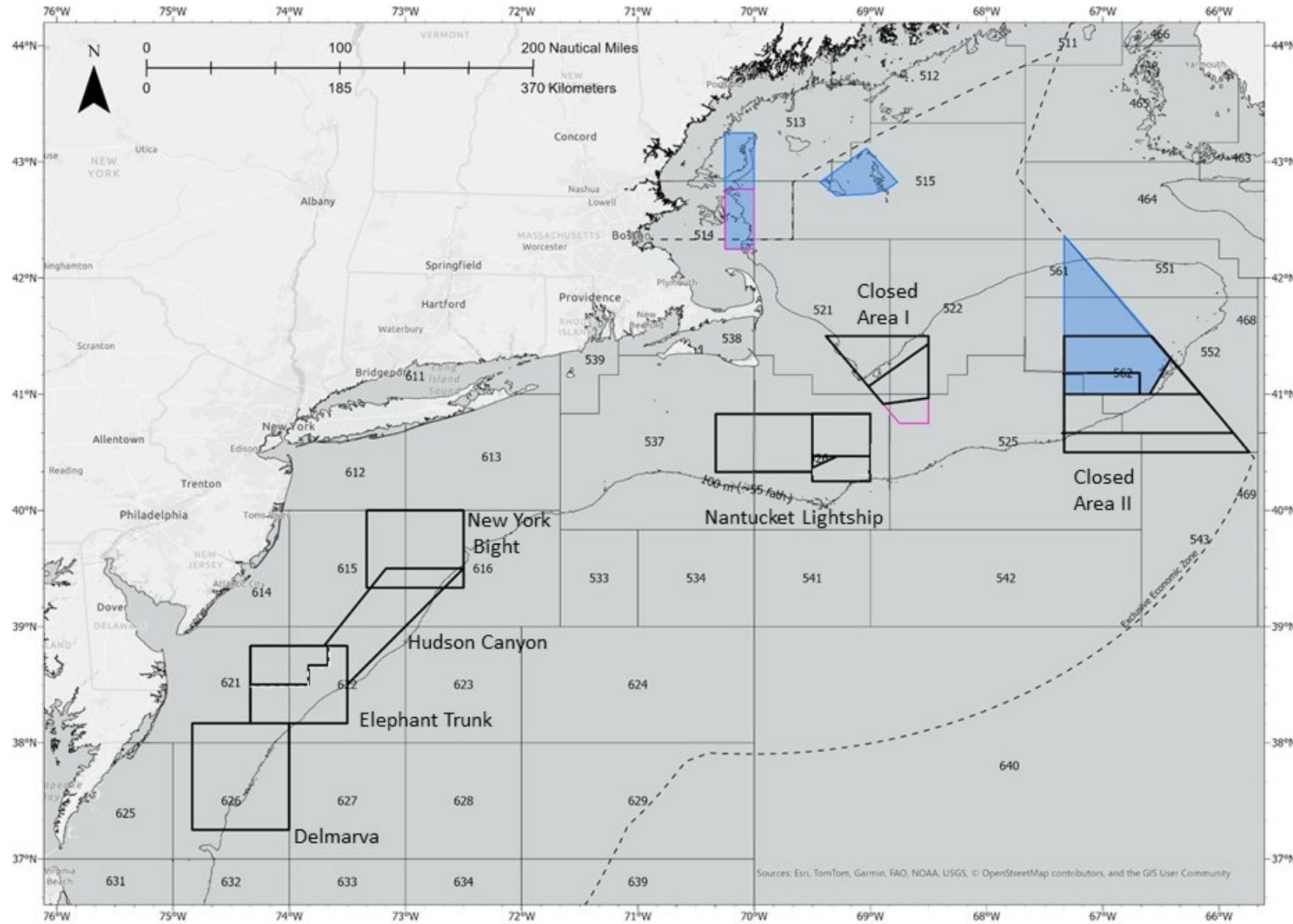
## Northern Gulf of Maine

- Special fishery with overall quota and 200 lb/day limit (derby fishery)



# Overview

Figure 2.1



HC: closed in 1998, substantial landings in 2001-2007, closed 2008, opened 2011, closed 2014.

ET: closed in 2004, ~30,000 mt landed after opened in 2007, closed 2012, reopened in 2015 (part) and 2017 (remainder)

DMV: closed 2007, opened 2009, closed 2012, reopened 2014

HC-ET-DMV combined into Mid-Atlantic Access Area in 2016

DMV back to open in 2018  
Remaining MAAA back to open in 2022

NY Bight new in 2022, closed

# Landings

## 1964-1981

- Port agents assigned area and gear fished to dealer landings
- Weight out of landings (WOLAND)

## 1982-1993

- Slight modification in data collection
- Weigh out detailed (WODET)

## 1994-2019

- Change in data collection, mandatory reporting
- Dealer reports (DR, no area fished) and Vessel Trip Reports (VTR, with area fished)
- Area Allocation (AA) method to match DR to VTR to assign area fished to landings
- Commercial fisheries detailed with area allocation (CFDET AA)

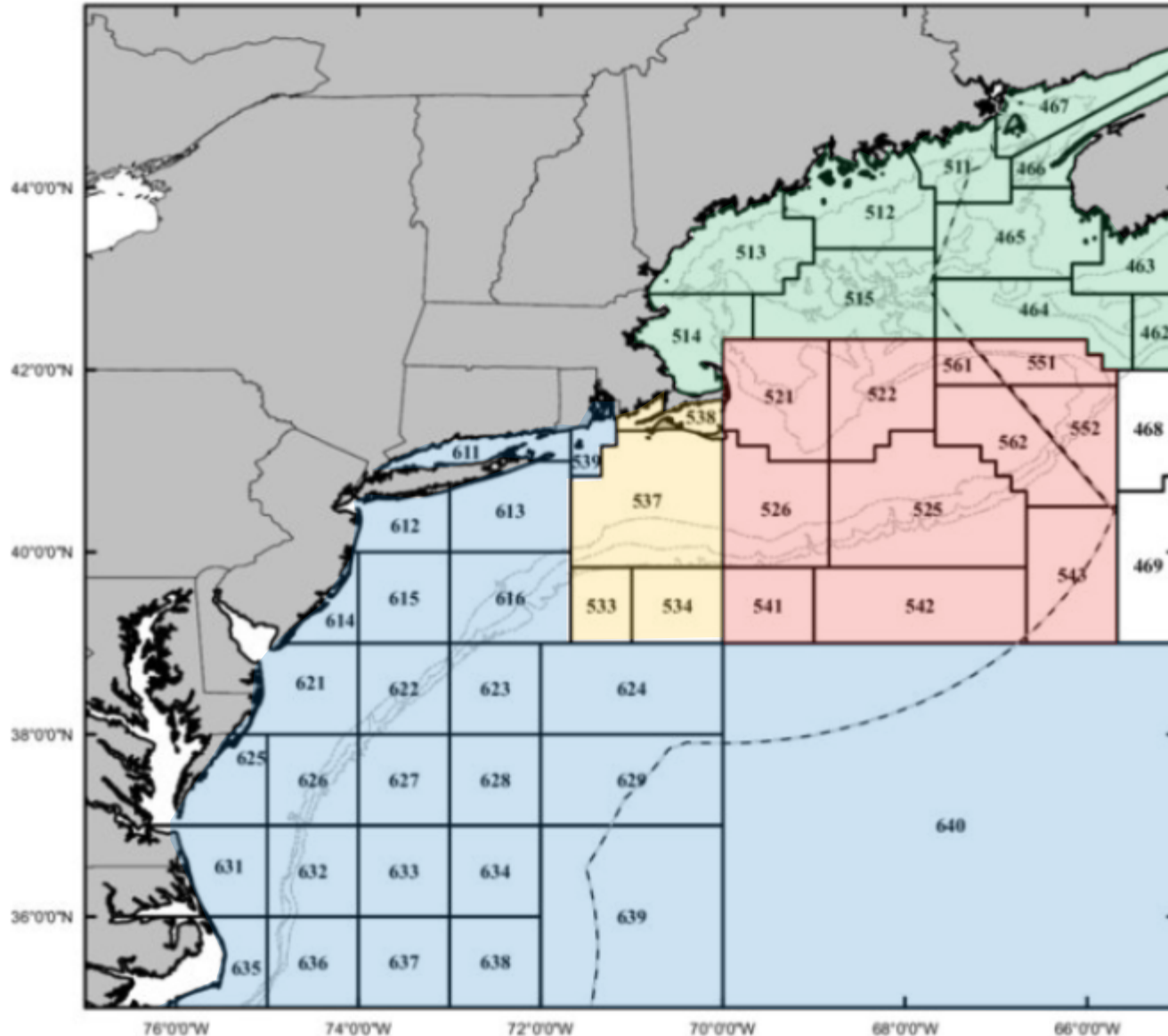
## 2000-present

- Single database for quota monitoring and stock assessments (DR, VTR, AA, discards)
- Catch Accounting and Monitoring System (CAMS)



<https://www.abc6.com/mayor-mitchell-advocates-for-scallopers/>

# Landings



## Areas

Mid-Atlantic (MA):  
≥ 600, and 539 (Block Island)

Southern New England (SNE):  
533, 534, 536-538

Georges Bank (GB):  
520-526, 541-543, 561, 562, 551, 552

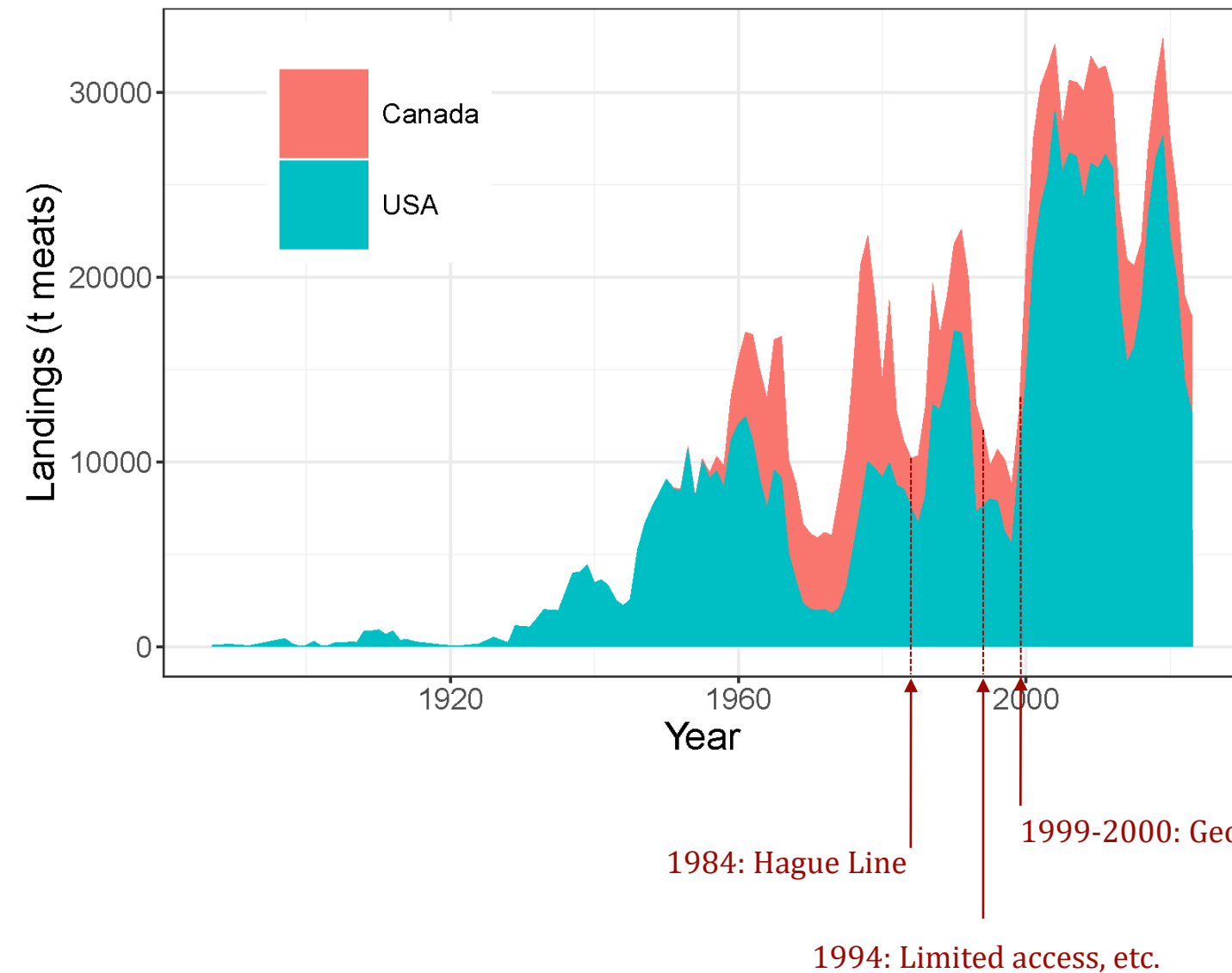
Gulf of Maine (GOM):  
>400, 500, 511, 512, 513, 514, 515

## Gears

Three gear groups:

- Trawls (otter and beam)
- Dredge (scallop, clam/quahog, rakes)
- Other (all other gears)

# Landings



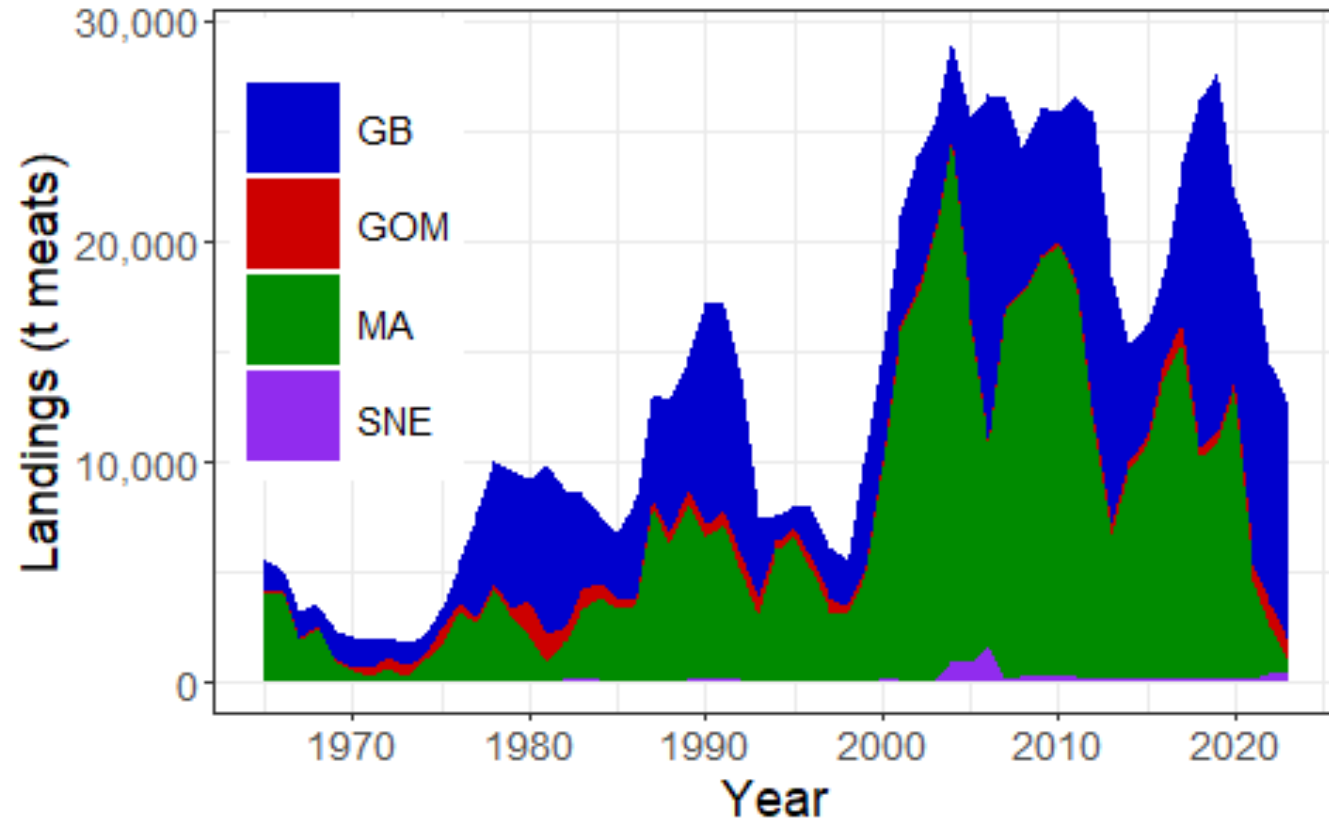
## Long-term landings

- U.S. and Canada, 1887–2023
- Prior to 1920: mostly ME, nearshore
- Dip during WW II
- Since 1960: boom and bust cycles
- Since 2001: high landings in most years
- Max US: 29,109 mt meats in 2004
- 2023: 12,699 mt



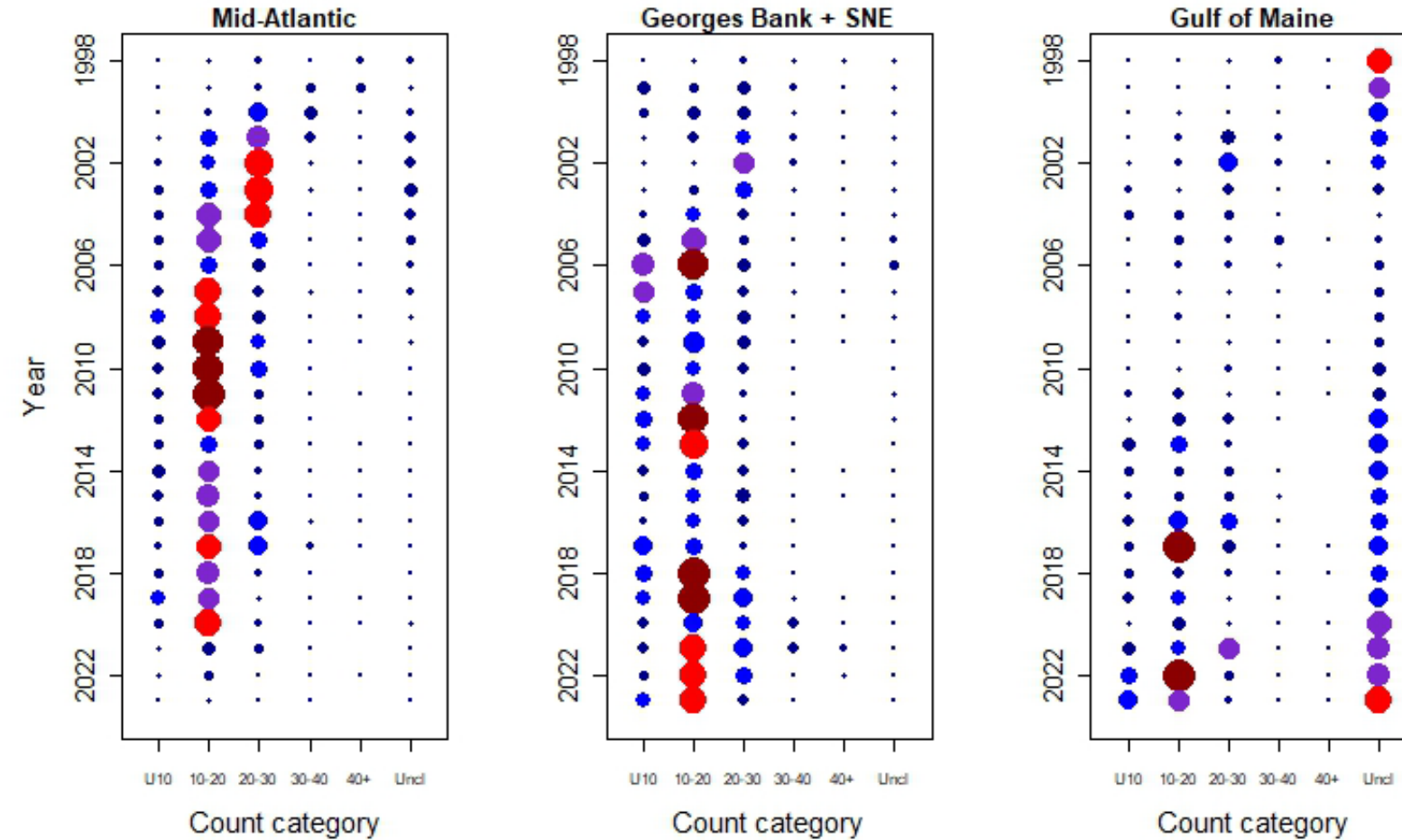
# Landings

## Landings by region



- **Mid-Atlantic:** increase in 1980s and early 2000, peak in 2004(23,565 mt), decline with poor 2007-2009 year classes, below average recruitment since 2013, under 700 mt in 2023
- **Georges Bank:** early peaks (1980s, 1990s), decline in 1993, stayed low until 2004 (~5,000 mt annually), big increases in 2005-2006 (opening of groundfish areas) and 2012-2013 (shift of effort from MA), peak in 2013 (16,367 mt), represents highest proportion of landings in recent years
- **GOM and SNE:** minor in comparison

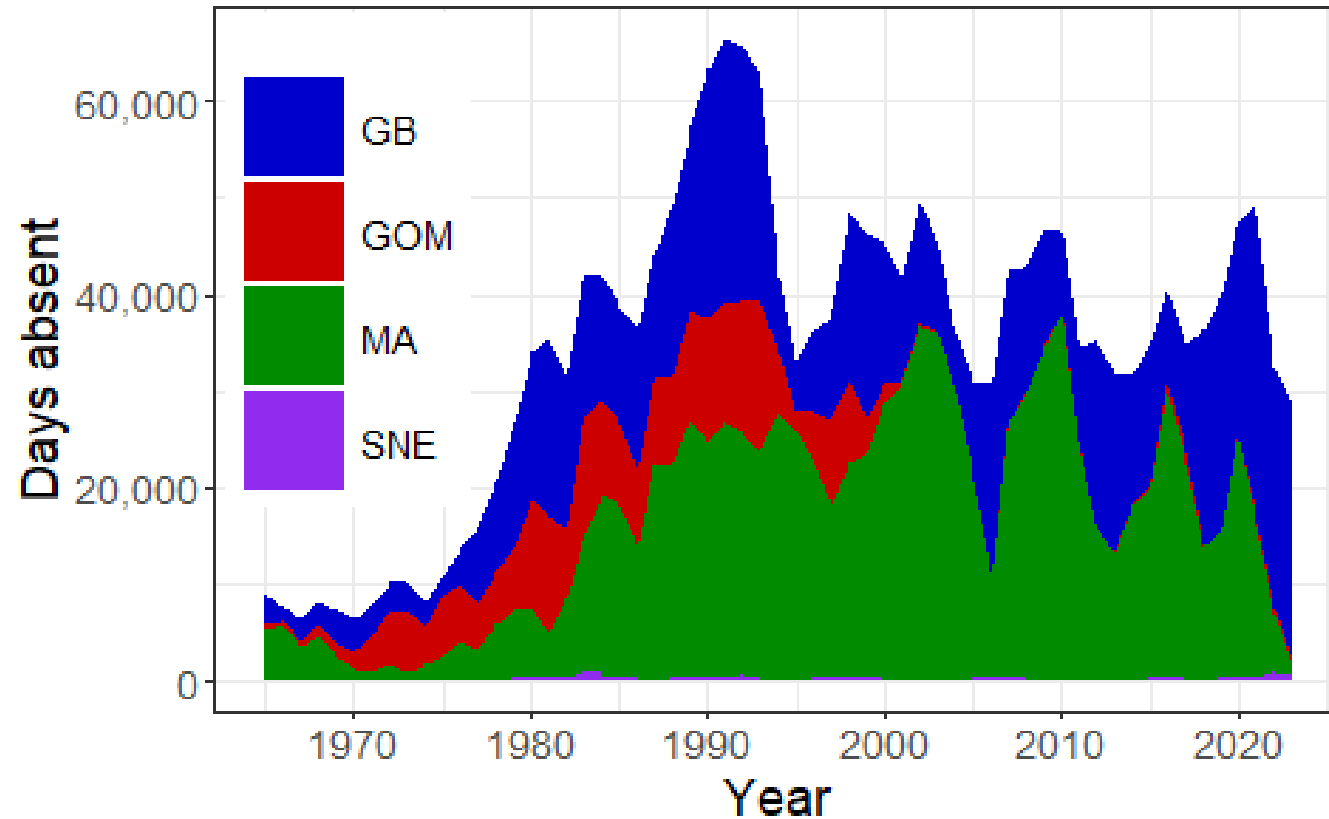
# Landings



## Landings by region and meat count

- 5 market categories (#/lb)
- Early years: mostly 20-30+ count
- Trend towards larger scallops
- Since early 2000: ~75% of landings are below 20 count
- Shift of large scallops (U10, >45g) from MA to GB
- GOM different due to lack of rotational areas, and scallops landed in shell (unclassified)

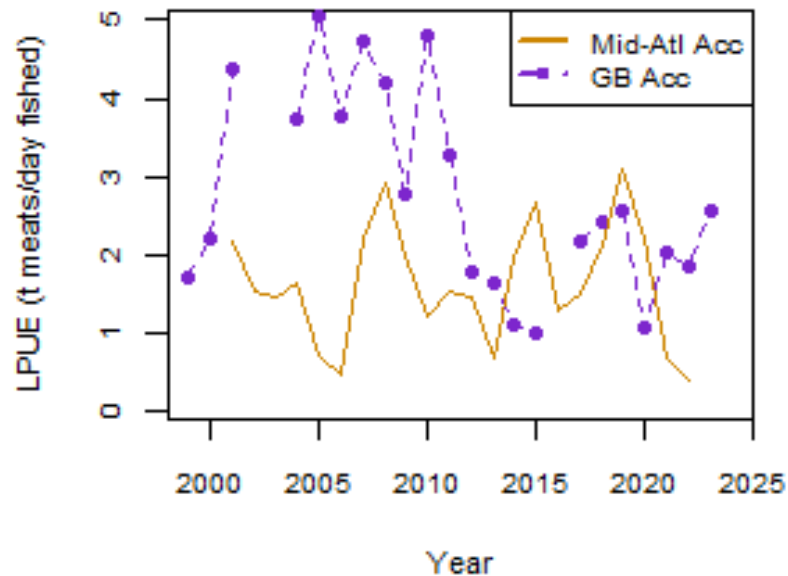
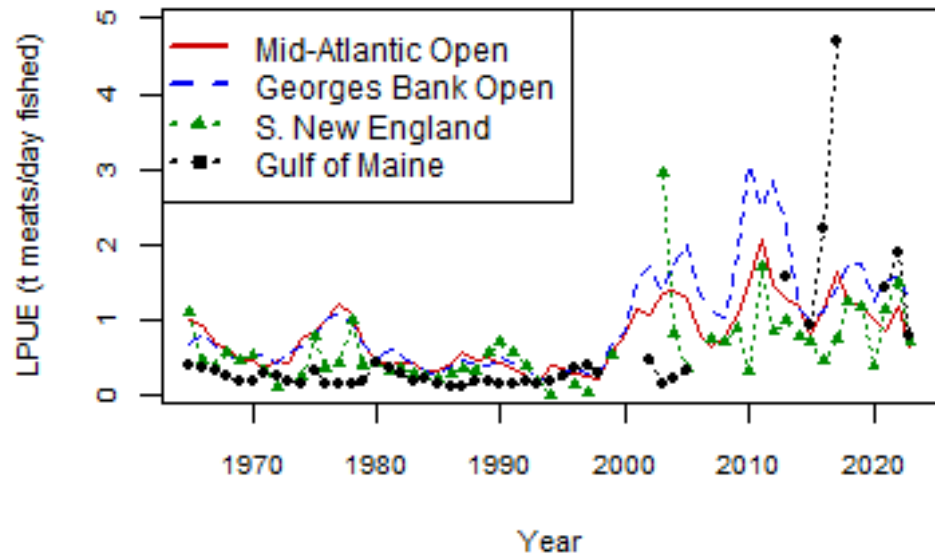
# Fishing Effort and LPUE



## Limited access effort (days absent)

- **Mid-Atlantic:** increasing importance in 1970s, peak in early 2000s, then decrease on average
- **Georges Bank:** increased importance as effort has shifted from MA
- **GOM:** significant effort prior to 2000 due to vessels targeting sea urchins and landing sea scallops
- Overall effort in SNE and GOM is minimal

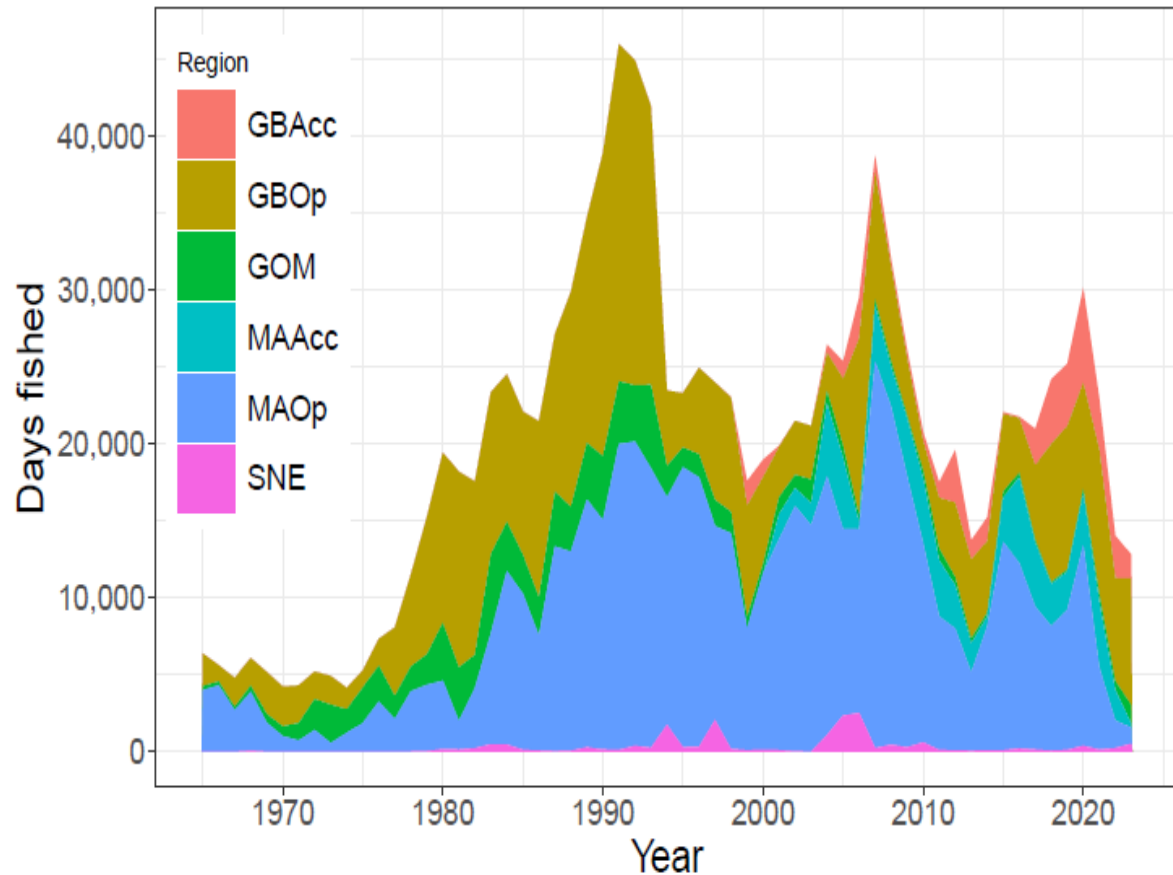
# Fishing Effort and LPUE



## Limited access LPUE

- Landings per days fished, where days fished = time the gear is fishing (from port interviews, observers since 1994)
- Open area (1965-2023, top)
- Access areas (1998-2023, bottom)
- See decreasing trend up to ~1998, increase since stock recovered (2001)





## Limited access effort (days fished)

- General increase from mid 1970s to 1991
- Decrease in 1990s due to low catch rates and then effort reduction measures
- Increase in MA (2000-2007) from reactivation and increase in gen cat
- Total effort since 2007 variable and decreasing

# Fishing Effort and LPUE

## Vessel Monitoring System fishing effort

- All fleets combined
- Fishing: 2.0-5.5 knots speed

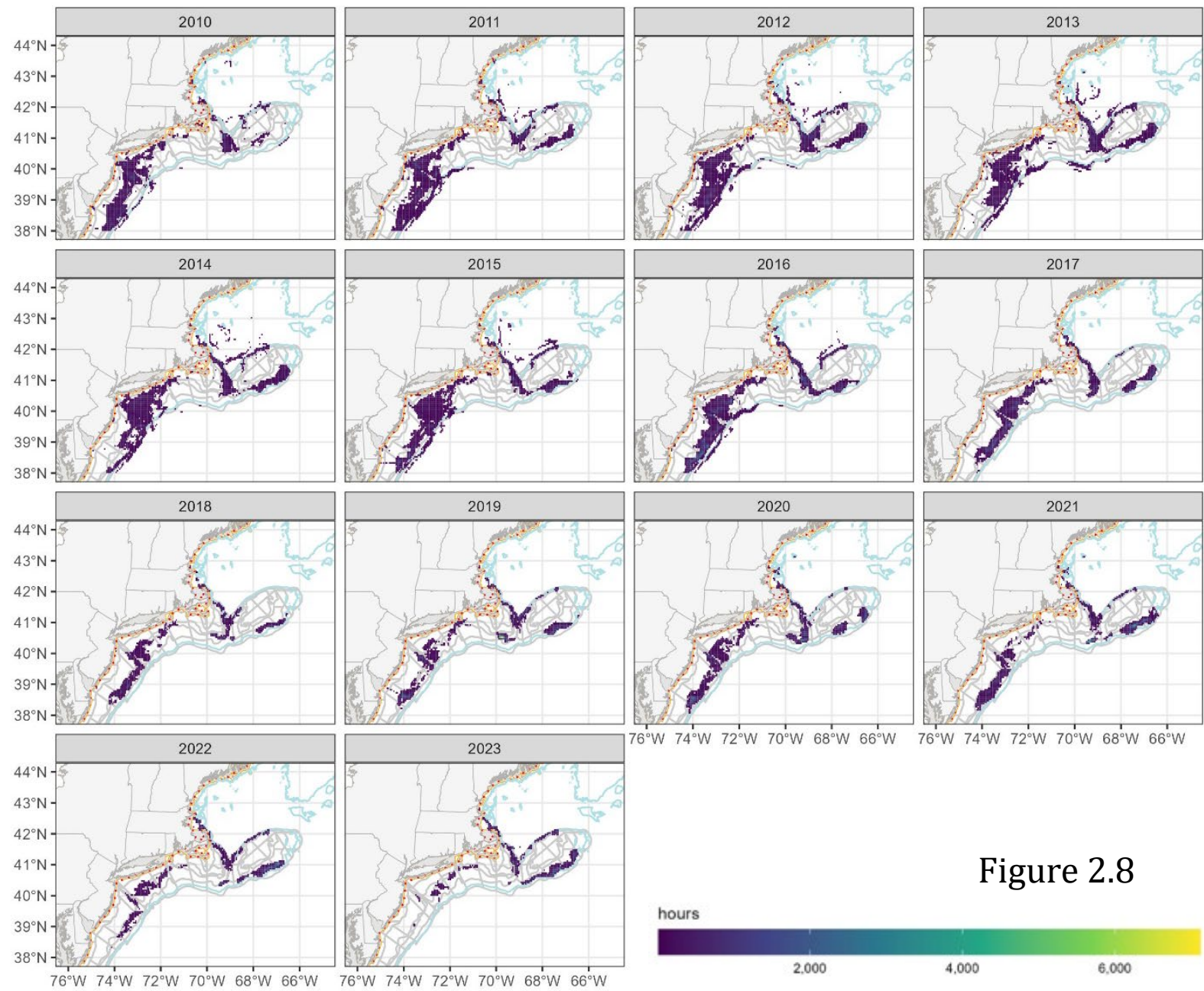


Figure 2.8

# Discards

Northeast Fishery Observer Program since 1989

- Sampling intensity on non-access trips was low until 2003

## Discards source:

1989-2009: SARC 59

- Using SBRM approach (Wigley et al., 2007)
- Ratio estimator (Cochran 1963) 
$$D = \frac{d}{k_{all}} * K$$

2010-2017:

- From SARC 65, using SBRM approach

2018-2023: CAMS

- Subtrip specific
- Uses d/kall ratio then applied to unobserved subtrips
- Discard mortality rate applied



Scallops are discarded due to:

- small size (i.e., not profitable)
- high-grading (e.g., access area trips)

May be mortality

- On deck (crushing, high temp, desiccation)
- Return to water (stress, shock, damage, predation)

Discard mortality estimated as 20%, but may be higher in MA in summer (high temps)

- Careful handling and quick return to water increases chance of survival

Discard proportion of landings:

- 5.2% for 1989-2023
- 5.9% for 2010-2023
- Variable by region (high SNE 2014-2020)

| Year | MA    |      |      | SNE  |      |      | GB    |      |      | GOM  |      |     | Total |      |
|------|-------|------|------|------|------|------|-------|------|------|------|------|-----|-------|------|
|      | Disc  | CV   | % L  | Disc | CV   | % L  | Disc  | CV   | % L  | Disc | CV   | % L | Disc  | CV   |
| 1989 | 213   |      | 2.7  |      |      |      | 4     |      | 0.1  |      |      |     | 217   |      |
| 1990 | 8     |      | 0.1  |      |      |      | 1     |      | 0.0  |      |      |     | 9     |      |
| 1991 | 11    |      | 0.2  |      |      |      | 5     |      | 0.1  |      |      |     | 16    |      |
| 1992 | 131   |      | 2.6  |      |      |      | 465   |      | 5.6  |      |      |     | 596   |      |
| 1993 | 22    |      | 0.8  |      |      |      | 346   |      | 9.4  |      |      |     | 368   |      |
| 1994 | 703   |      | 11.8 |      |      |      | 10    |      | 0.8  |      |      |     | 713   |      |
| 1995 | 495   |      | 7.5  |      |      |      | 23    |      | 2.3  |      |      |     | 518   |      |
| 1996 | 38    |      | 0.7  |      |      |      | 116   |      | 5.4  |      |      |     | 154   |      |
| 1997 | 11    |      | 0.4  |      |      |      | 46    |      | 2.0  |      |      |     | 57    |      |
| 1998 | 53    |      | 1.8  |      |      |      | 4     |      | 0.2  |      |      |     | 57    |      |
| 1999 | 114   |      | 2.5  |      |      |      | 142   |      | 2.7  |      |      |     | 256   |      |
| 2000 | 813   |      | 8.7  |      |      |      | 991   |      | 20.0 |      |      |     | 1,804 |      |
| 2001 | 1,969 |      | 12.6 |      |      |      | 531   |      | 10.8 |      |      |     | 2,500 |      |
| 2002 | 1,970 |      | 11.3 |      |      |      | 107   |      | 1.8  |      |      |     | 2,077 |      |
| 2003 | 2,244 |      | 11.1 |      |      |      | 332   |      | 6.8  |      |      |     | 2,576 |      |
| 2004 | 2,559 |      | 10.9 |      |      |      | 102   |      | 2.3  |      |      |     | 2,661 |      |
| 2005 | 424   |      | 2.7  |      |      |      | 238   |      | 2.7  |      |      |     | 662   |      |
| 2006 | 244   |      | 2.6  |      |      |      | 378   |      | 2.4  |      |      |     | 622   |      |
| 2007 | 294   |      | 1.8  |      |      |      | 236   |      | 2.5  |      |      |     | 530   |      |
| 2008 | 457   |      | 2.6  |      |      |      | 341   |      | 5.3  |      |      |     | 798   |      |
| 2009 | 1,013 |      | 5.3  |      |      |      | 398   |      | 5.8  |      |      |     | 1,402 |      |
| 2010 | 730   | 0.18 | 3.7  | 2    | 1.18 | 1.0  | 715   | 0.17 | 12.1 | 3    | 0.67 | 1.8 | 1,450 | 0.12 |
| 2011 | 537   | 0.14 | 3.0  | 19   | 0.44 | 8.6  | 536   | 0.21 | 6.5  | 2    | 0    | 0.8 | 1,094 | 0.13 |
| 2012 | 278   | 0.10 | 2.4  | 0    | -    | 0.0  | 890   | 0.10 | 6.5  | 0    | -    | 0.0 | 1,168 | 0.08 |
| 2013 | 157   | 0.17 | 2.4  | 5    | 0.33 | 6.2  | 357   | 0.14 | 3.1  | 6    | 0.31 | 1.2 | 525   | 0.11 |
| 2014 | 70    | 0.16 | 0.7  | 20   | 0.47 | 21.9 | 220   | 0.16 | 4.2  | 9    | 0.40 | 2.4 | 319   | 0.12 |
| 2015 | 562   | 0.11 | 5.3  | 11   | 0.45 | 13.9 | 191   | 0.18 | 3.7  | 2    | 0.21 | 0.5 | 766   | 0.09 |
| 2016 | 2,074 | 0.07 | 15.1 | 18   | 0.24 | 17.6 | 98    | 0.20 | 2.4  | 6    | 0.30 | 0.9 | 2,190 | 0.07 |
| 2017 | 931   | 0.22 | 6.0  | 3    | 0.23 | 2.6  | 479   | 0.11 | 6.9  | 34   | 0.22 | 3.6 | 1,447 | 0.15 |
| 2018 | 429   | 0.15 | 4.2  | 22   | 0.34 | 20.6 | 852   | 0.15 | 5.4  | 10   | 0.28 | 2.6 | 1,313 | 0.11 |
| 2019 | 461   | 0.12 | 4.3  | 37   | 0.34 | 23.3 | 1,235 | 0.12 | 7.5  | 2    | 0.30 | 0.5 | 1,735 | 0.09 |
| 2020 | 678   | 0.19 | 5.2  | 44   | 0.77 | 27.2 | 1,059 | 0.30 | 12.3 | 2    | 0.73 | 0.4 | 1,782 | 0.19 |
| 2021 | 272   | 0.21 | 5.9  | 25   | 0.52 | 14.7 | 1,427 | 0.13 | 10.1 | 17   | 0.34 | 2.2 | 1,741 | 0.11 |
| 2022 | 106   | 0.24 | 5.1  | 26   | 0.47 | 6.9  | 519   | 0.15 | 4.8  | 54   | 0.18 | 4.9 | 704   | 0.12 |
| 2023 | 78    | 0.92 | 11.7 | 59   | 1.12 | 15.8 | 728   | 0.22 | 6.8  | 51   | 0.17 | 5.5 | 916   | 0.20 |



# Incidental mortality

Scallops not caught, but damaged by the dredge (e.g., crushing)

## **Estimates of incidental mortality** (% of scallops in track of dredge that die)

- Caddy (1973): 15-20%, hard bottom in Canada
- Murawski & Serchuk (1989): 5%, sandy bottom in New Jersey
- Ferraro et al. (2017): 2.5% for MA, 8% for GB



Ratio  $R$  of scallops in the path of a dredge that were caught, to those killed but not caught:

$$R = \frac{e}{c(1 - e)}$$

$c$  is the fraction of scallops that die among scallops in the path of the dredge, and  $e$  is efficiency of the dredge.

Rate of indirect (non-landed) fishing mortality (Hart 2003):  $F_I = F_L / R = \frac{1}{e} F_L c (1 - e)$

Assuming scallop are fully selected to the gear:

CASA model: assume  $c = 0.06$  in MA and  $c = 0.11$  for GB

SYM and SAMS model: assume  $c = 0.05$  in MA and  $c = 0.1$  for GB

Discarding and discard mortality

- Not modeled by CASA model
- Included in SYM and SAMS model



# Commercial shell height data

From observer data

- Kept scallops since 1992
- Discarded scallops since 1994

Interpret with caution for years prior to 2003  
(limited coverage in open areas)

Observer coverage rate >5% since 2003,  
> 10% in some years

Scallops between 70-90 mm often made up a  
considerable portion of the landings during  
1975-1998

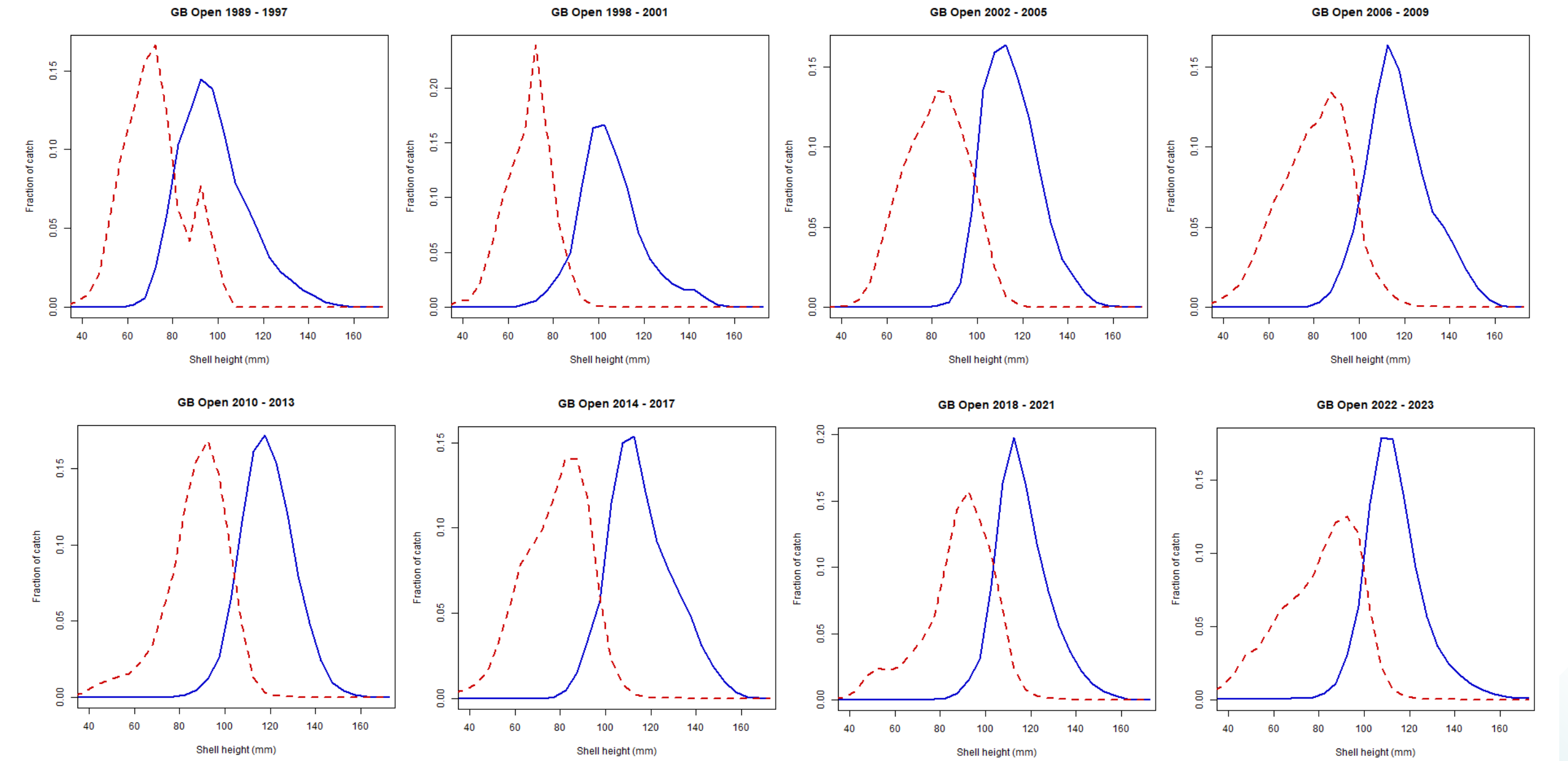
Sizes selected by the fishery have increased,  
scallops less than 90 mm were rarely taken  
since 2002



# Commercial shell height data

Georges Bank open areas

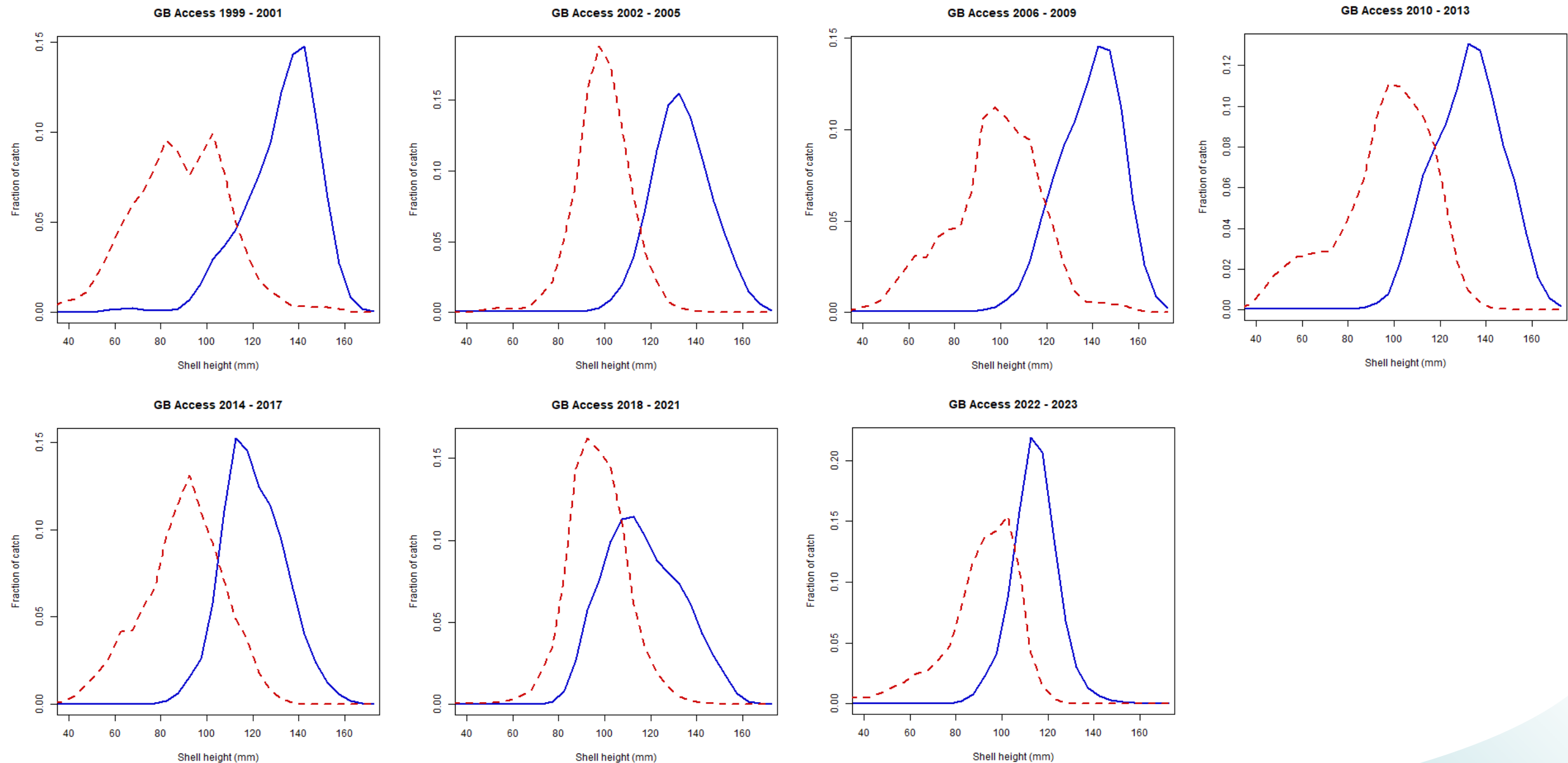
Figure 2.9 (a)

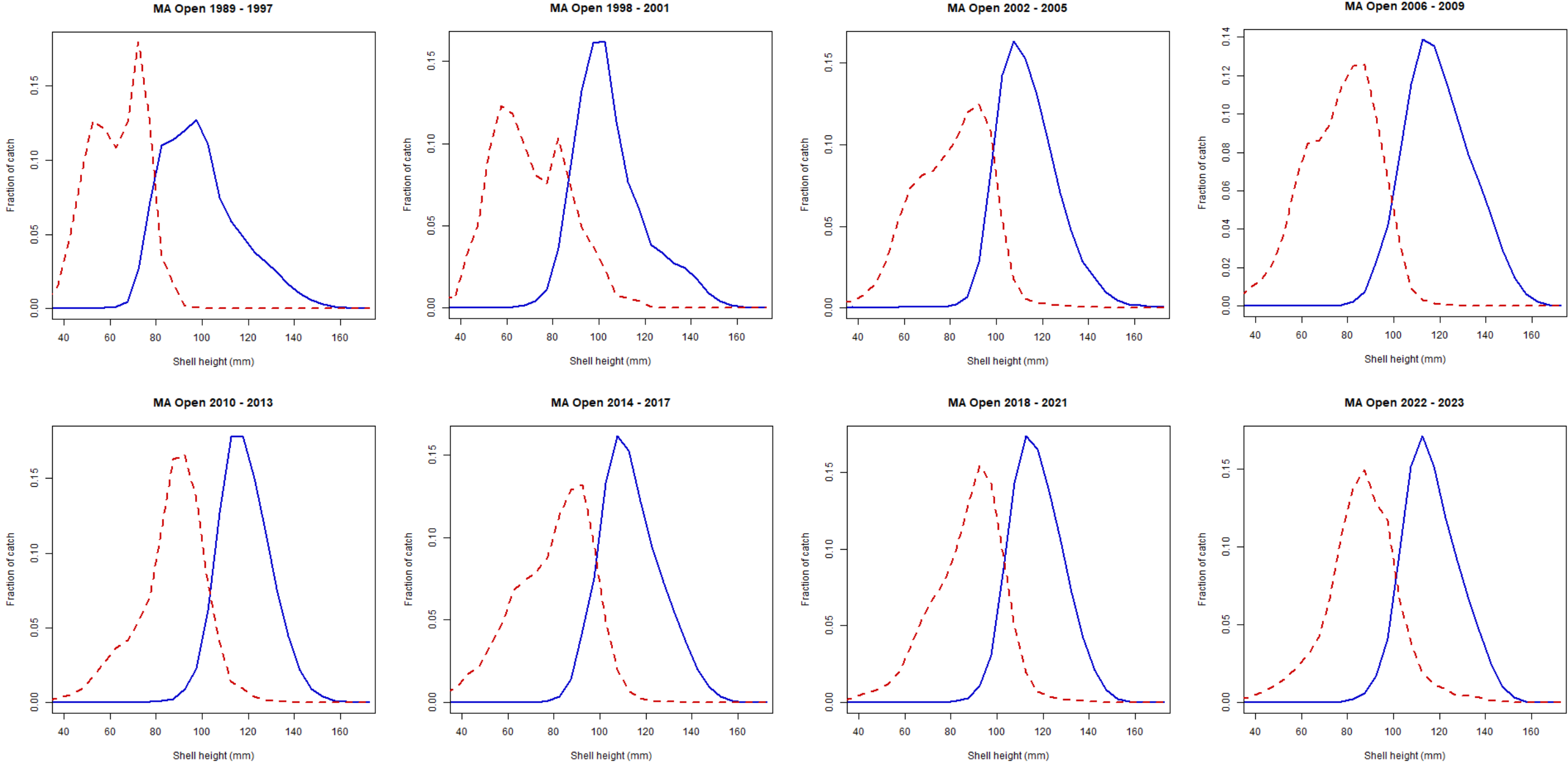


# Commercial shell height data

Georges Bank access areas

Figure 2.9 (b)



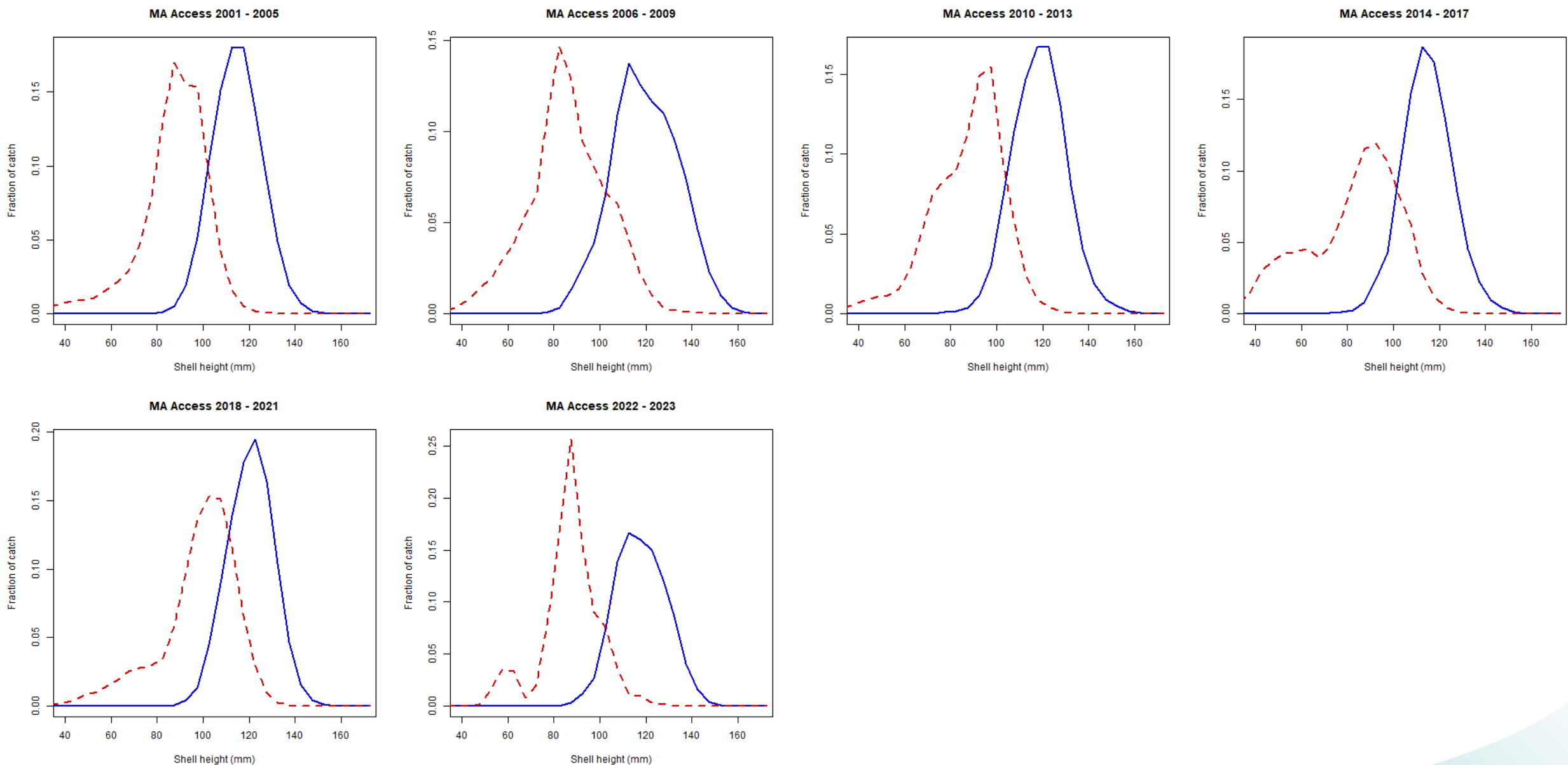




# Commercial shell height data

Mid-Atlantic access areas

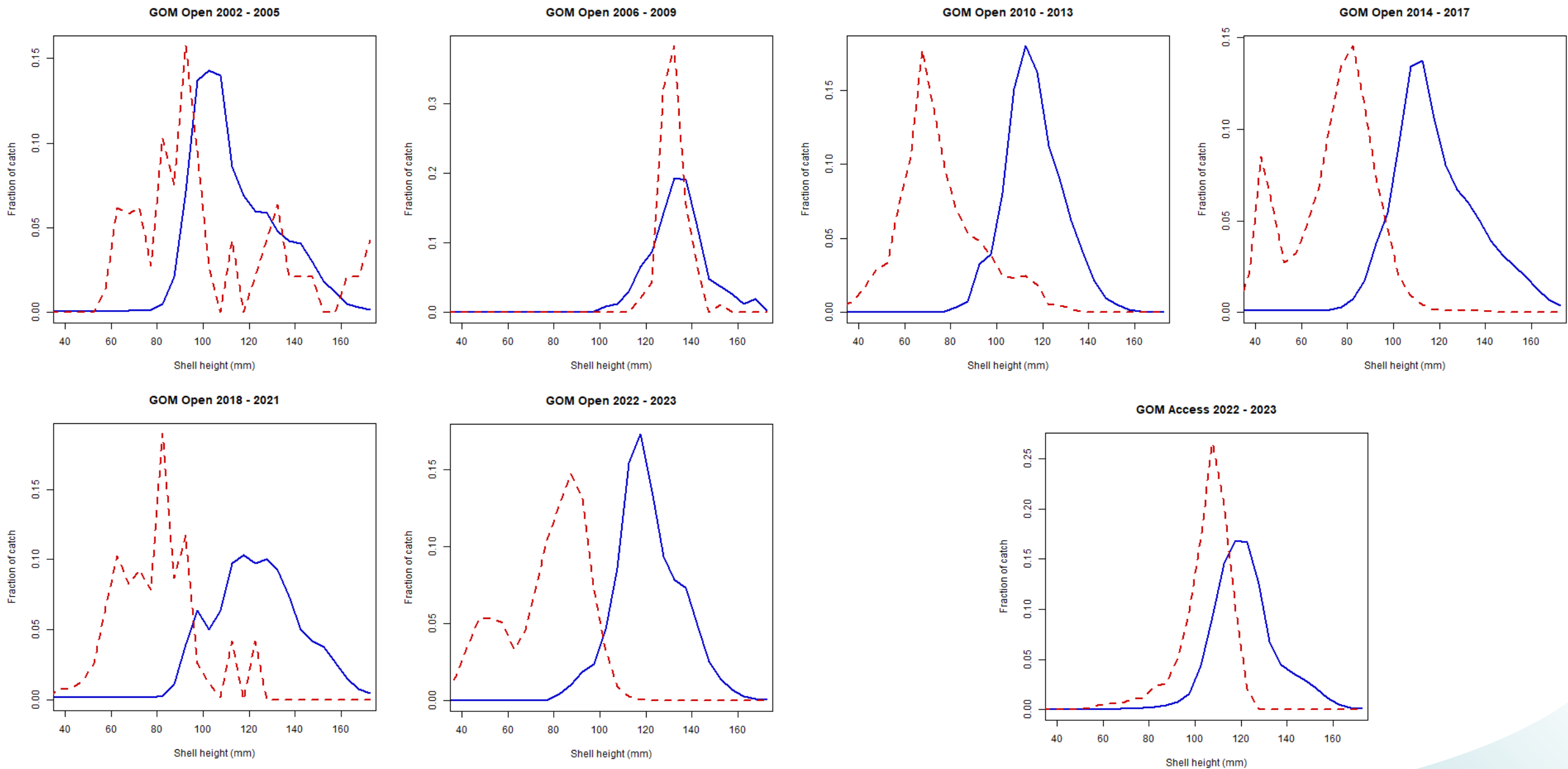
Figure 2.9 (d)



# Commercial shell height data

## Gulf of Maine open areas and NGOM

Figure 2.9 (e)



# Thank you

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